

1. TARGETED FIELD QUESTIONNAIRE MATRIX (32 QUESTIONS)

Category 1: Commuters & Daily Transit Users (Mobility Lens \$S_m\$ & Telemetry)

Q#	SURVEY QUESTION	TECHNICAL TARGET / AI SYSTEM MAPPING
C1.1	What primary transport mode do you use across the Bhopal–Sehore–Ashta corridor (BRTS, city bus, private vehicle, two-wheeler)?	Classifies telemetry weighting based on commuter mode density.
C1.2	What is your average transit delay (in minutes) at known chokepoints (Karond Circle, Sehore Naka, Ashta Bypass) during peak hours (8–10 AM, 5–7 PM)?	Calibrates S_m transit delay normalization baseline: $S_m = \min(10.0, (\text{delay_mins} / 15.0) \cdot 10.0)$.
C1.3	By how many minutes does your travel time increase when moderate-to-heavy rainfall occurs along State Highway 18?	Maps cross-lens interaction terms between Climate (S_c) and Mobility (S_m).
C1.4	When major underpasses or roads flood, do standard navigation apps reroute you accurately, or do you rely on informal community alerts?	Evaluates real-time routing gaps for citizen WhatsApp crowdsourcing integration.
C1.5	How frequently are public transit routes canceled or diverted during waterlogging without prior digital notification?	Establishes GTFS-RT feed update frequency requirements for the FastAPI ingestion pipeline.
C1.6	If you encounter a severe road hazard, would you submit a 1-click GPS drop and photo via WhatsApp if available?	Sets target user payload schemas for <code>POST /api/v1/telemetry/citizen</code> .
C1.7	What automated feedback (e.g., "Report Verified", "Crew En Route") would increase your trust in reporting urban hazards?	Designs real-time feedback event triggers in the Server-Sent Events (SSE) bus.

Category 2: Municipal Field Crews & First Responders (Locks, Offline PWA & Escalation)

Q#	SURVEY QUESTION	TECHNICAL TARGET / AI SYSTEM MAPPING
F2.1	What channel currently alerts you to urgent field incidents (phone call, SMS, WhatsApp group, or official portal)?	Benchmarks legacy notification latency against the Gemini Flash automated action plan pipeline.
F2.2	How often do multiple field units arrive at the same incident location due to uncoordinated dispatching?	Quantifies the baseline loss that the 60-second Firestore TTL lock mechanism eliminates.
F2.3	What exact geographic pockets along the Bhopal–Sehore–Ashta route experience complete cellular signal loss (2G/No Service)?	Maps offline zone coordinates to configure Workbox Service Worker sync triggers.
F2.4	How do you record incident field notes, photos, and status updates when operating entirely offline?	Defines IndexedDB storage schemas (<code>field_outbox</code>) for local PWA state caching.
F2.5	How long does it typically take to regain connectivity and sync field updates back to the central ward office?	Sets retry policy and backoff intervals for <code>POST /api/v1/field/sync</code> .
F2.6	What is your realistic target response time (in minutes) from receiving a high-priority alert to deploying pump trucks or barriers on-site?	Tunes escalation thresholds for the automated 15-minute SMS monitor.
F2.7	What mobile interface elements (large high-contrast buttons, voice-to-text, dark mode) are necessary when working in heavy rain or sunlight?	Directs UI component layout for the SCR-04 Field Mobile PWA.
F2.8	If a critical incident remains unhandled for over 15 minutes, who in your organizational chain of command must receive an immediate escalation?	Configures recipient fallback logic for Twilio SMS escalation webhooks.

Category 3: Ward Engineers & Municipal Admins (Climate Lens \$S_c\$, Weights & AI Guardrails)

Q#	SURVEY QUESTION	TECHNICAL TARGET / AI SYSTEM MAPPING
A3.1	What maximum rainfall intensity (mm/hr) can your ward's storm-water drainage network process before surface runoff begins?	Establishes the S_c lens baseline formula: $S_c = \min(10.0, (\text{rain_mm_hr} / 50.0) \cdot 10.0)$.
A3.2	Are storm drains cleared reactively after flooding occurs or proactively based on weather forecast alerts?	Determines the necessity for predictive Gemini 2.5 Flash resource allocation plans.
A3.3	What live weather or hydrological data streams (IMD sensors, manual gauges) are currently accessible via REST APIs or open feeds?	Identifies external ingestion endpoints for the FastAPI serverless gateway.
A3.4	Which factor poses the highest operational risk in your jurisdiction: traffic gridlock (S_m), heavy rain (S_c), or vulnerable populations (S_v)?	Establishes default baseline weights for $R_c = (W_m \cdot S_m) + (W_c \cdot S_c) + (W_v \cdot S_v)$.

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A3.5	Would you support shifting risk weights dynamically depending on the season (e.g., higher \$W_c\$ during monsoons, higher \$W_m\$ during festival rush)?	Validates the requirement for the SCR-05 Admin Weight Tuning Console.
A3.6	What validation guardrails must an AI dispatch draft pass before it can be sent to field crews?	Configures JSON schema validation and prompt system instructions for Gemini 2.5 Flash.
A3.7	Is a ~461-meter hexagon (Uber H3 Res-8) granular enough to pinpoint incidents, or do dense urban cores require ~160-meter cells (Res-9)?	Tests spatial index scaling parameters in the backend/app/engine/ module.

Category 4: Local Residents & Vulnerable Communities (Vulnerability Lens \$S_v\$ & Access)

Q#	SURVEY QUESTION	TECHNICAL TARGET / AI SYSTEM MAPPING
R4.1	In informal settlements or older urban wards (e.g., Old Bhopal), what physical lane widths (<3 meters) prevent emergency vehicle access?	Sets spatial vulnerability penalties (\$S_v\$) based on access limitations.
R4.2	What proportion of households in your neighborhood experience power outages or water contamination during minor storm events?	Provides demographic sensitivity data to calibrate static \$S_v\$ overlays.
R4.3	What primary factors cause local drainage blockages (uncollected plastic waste, construction debris, undersized culverts)?	Identifies contextual tags to pass into Gemini Flash prompt context.
R4.4	How do residents in your area currently receive urgent flooding or disaster warnings (sirens, public loudspeakers, SMS, community leaders)?	Assesses integration paths for multi-channel alert delivery.
R4.5	What is the maximum historical flood depth (in feet/meters) recorded inside buildings in your area over the last 3 monsoon cycles?	Validates historical flood level layers against H3 cell indices.
R4.6	How far (in meters or minutes) must residents travel during severe weather to reach the nearest operational shelter or clinic?	Calculates proximity risk factors for vulnerability scoring (\$S_v\$).

Category 5: Inter-City Highway Corridor (Bhopal – Sehore – Ashta Cross-Jurisdiction)

Q#	SURVEY QUESTION	TECHNICAL TARGET / AI SYSTEM MAPPING
H5.1	What exact highway markers or river bridge crossings on SH-18 experience complete traffic stoppage during sudden downpours?	Defines critical H3 cell clusters for regional telemetry tracking.
H5.2	How does long-haul freight traffic interact with emergency vehicle movement along the inter-city corridor during disruptions?	Refines multi-lens cross-talk logic when both \$S_m \ge 8.0\$ and \$S_c \ge 8.0\$.
H5.3	How do Bhopal municipal authorities share real-time incident data with Sehore and Ashta district officials during cross-boundary events?	Proves the necessity for multi-tenant jurisdiction filters in SCR-01/SCR-02.
H5.4	Are key power substations, water pumping stations, or telecom towers along the corridor located in low-lying, flood-prone zones?	Maps critical infrastructure points as high-priority nodes within \$S_v\$ calculations.

2. TECHNICAL NOVELTY ANGLES TO VALIDATE ON-GROUND

- H3 Spatial Resolution Granularity (Res-8 vs. Res-9):** Verify whether a ~461m hexagon (Res-8) accurately bounds local flood spots (e.g., Karond Circle in Bhopal, Ashta Bus Stand) or if high-density areas need dynamic resolution upsampling.
- Citizen Telemetry Reliability:** Test how fast citizen WhatsApp location drops can be validated against official OpenWeather and GTFS transit feeds without high false-positive rates.
- Semi-Urban Corridor Blind Spots:** Map network dead zones along State Highway 18 (Bhopal–Sehore–Ashta route) to establish realistic offline sync parameters for field crews.

3. FIELD PHOTOGRAPHY & DATA CAPTURE CHECKLIST (ENABLE GPS TAGS)

Mobility Lens (\$S_m\$) Artifacts

- Transit Bottlenecks:** Narrow chokepoints, broken signals, or illegal parking.
- Corridor Transitions:** Intersections where municipal transit intersects highway traffic.
- Infrastructure:** Digital route boards, bus stop conditions, vehicle crowding.

Climate & Hydrology Lens (\$S_c\$) Artifacts

- Drainage Infrastructure:** Clogged storm drains, missing gully covers, blocked outlets.
- High-Water Identifiers:** Watermarks on buildings, bridge piers, roadside walls.
- Topographical Runoffs:** Low-lying roads, underpasses, or dip sections.

Vulnerability Lens (\$S_v\$) Artifacts

- **Emergency Access Limits:** Narrow lanes (<3 meters) in informal settlements.
- **Infrastructure Exposure:** Power transformers, water pumps in flood-prone areas.
- **Response Nodes:** Municipal ward offices, police outposts, clinics.

Edge & Network Infrastructure

- **Connectivity Dead Spots:** Screenshots of network speed tests and offline app state.
- **Route Dead Zones:** Log exact mileage markers on SH-18 where cellular signal drops.